

Agentic Sales Orchestrator

Programmatic Lead Table Extension Runbook

Enterprise runbook for programmatic Dataverse metadata creation on the existing Lead table

Item	Value
Version	v1.0 - 2026-05-23
Target environment	Phoenicarix-CI - https://phoenicarix-ci.crm4.dynamics.com
Solution	ASO.Core - unique name ASOCore
Target table	Lead - logical name lead
Implementation method	.NET 8 console application using Dataverse metadata requests
Prepared for	Customer IT team, Product Owner, Dynamics 365 / Power Platform developers

Purpose: Explain what was done, why it was done programmatically, how the script works, how to validate it, and how customer IT can maintain or repeat the pattern safely.

1. Executive summary

We extended the existing Microsoft Dataverse Lead table programmatically instead of creating every custom column manually in the Power Apps maker UI. The target was the Agentic Sales Orchestrator Phase 2 data model in ASO.Core. The script connected to the Phoenicarix-CI Dataverse environment, checked whether each `aso_Lead` column already existed, created missing columns through Dataverse metadata requests, associated the created columns with the ASO.Core solution, and published the Lead table customizations. In simple terms: instead of clicking New column more than thirty times, we gave Dataverse a controlled, repeatable instruction list.

2. Why we used a script instead of Copilot or manual UI

Copilot was useful for discussion and planning, but it could not reliably execute changes against the existing standard Lead table from the Create new tables canvas. Manual UI creation is safe but slow and error-prone for this number of fields. A metadata script gives a faster, repeatable, auditable, and solution-aware approach.

Option	What happened / reason	Decision
Copilot in Create new tables workspace	Could not reliably modify the existing standard Lead table and returned errors.	Not used for implementation.
Manual UI column creation	Safe, but slow and error-prone for more than thirty fields.	Avoided where possible.
Programmatic metadata creation	Fast, repeatable, validates existing columns, keeps schema names consistent, and can target a solution.	Chosen for Lead extension.

3. Explanation for different audiences

Audience	Explanation
Grandma	We added many new labeled drawers to the Lead record. Instead of adding each drawer by hand, we used a small program to add them correctly.
Product Owner	We created the data foundation needed for AI qualification, Foundry orchestration, SAP context, Customer Insights governance, and sales-agent output tracking on Leads.
Developer	We used a .NET 8 console application with <code>Microsoft.PowerPlatform.Dataverse.Client</code> and Dataverse metadata requests, especially <code>CreateAttributeRequest</code> , to create Lead attributes in ASO.Core.
Admin / Operations	The script is repeatable and skips existing fields, so it can be safely rerun after partial execution, then validated in ASO.Core -> Tables -> Lead -> Columns.

4. Scope and non-scope

In scope	Not in scope
Connect to Phoenicarix-CI.	No HubSpot integration.
Target ASO.Core and the existing Lead table.	No SAP direct connection.

In scope	Not in scope
Create missing aso_Lead columns.	No Customer Insights journey activation.
Reuse existing global choices where defined.	No outbound customer email sending.
Create local choices where values belong only to Lead.	No Foundry orchestration call.
Publish Lead table metadata.	No form layout changes yet.

5. Prerequisites

Prerequisite	Status / target
Environment	Phoenicarix-CI
Dataverse URL	https://phoenicarix-ci.crm4.dynamics.com
Solution	ASO.Core
Solution unique name	ASOCore
Publisher prefix	aso
Target table	lead
Global choices	Created before running the Lead extension script
Local machine	macOS with .NET 8 installed
Authentication	Interactive OAuth login with the Power Apps user

6. Global choices used by the Lead extension

Several Lead columns reference global choices created earlier in ASO.Core. Dataverse may display names in lowercase internally; the script uses the lowercase logical names.

Global choice logical name used by script	Used by Lead column	Values / purpose
aso_aistatus	AI Agent Status	NotStarted, Running, Completed, Escalated, Failed, PartiallyCompleted, Blocked
aso_communicationstate	Communication State	NotEligible, Eligible, Queued, InJourney, WaitingInteraction, Completed, Suppressed, Failed, Blocked
aso_lifecyclecommunicationstage	Lifecycle Communication Stage	Lead, QualifiedLead, Opportunity, QuoteProposal, Order, Onboarding, Retention, Expansion
aso_journeyparticipationstatus	Journey Participation Status	NotStarted, Active, Paused, Completed, Exited, Error
aso_emailconsentstatus	Email Consent Status	OptedIn, OptedOut, Pending, Unknown, Blocked

7. Lead fields created programmatically

The following table is the implementation inventory for the Lead table extension. It can be copied into the project data dictionary or build tracker.

Display name	Schema/logical name	Dataverse type	Script metadata class	Notes
HubSpot Contact ID	aso_hubsportcontactid	Text	StringAttributeMetadata	External ingress key; HubSpot not configured yet.
HubSpot Source	aso_hubsportsource	Text	StringAttributeMetadata	Ingress source reference; kept as text for MVP flexibility.
SAP Business Partner ID	aso_sapbusinesspartnerid	Text	StringAttributeMetadata	SAP reference only.
SAP Customer ID	aso_sapcustomerid	Text	StringAttributeMetadata	SAP customer reference only.
AI Fit Level	aso_aifitlevel	Choice	Local PicklistAttributeMetadata	Local values: Strong, Moderate, Weak.
AI Lead Score	aso_aileadscore	Whole number	IntegerAttributeMetadata	Range 0-100.
AI Qualification Rationale	aso_aiqualificationrationale	Multiline text	MemoAttributeMetadata	Seller-facing AI rationale.
AI Outreach Draft	aso_aioutreachdraft	Multiline text	MemoAttributeMetadata	Draft only; not direct-send.
AI Routing Decision	aso_airoutingdecision	Choice	Local PicklistAttributeMetadata	Local values: Nurture, SDR, AE, Reject, NeedsReview, ExistingAccountReview.
AI Confidence	aso_aiconfidence	Decimal	DecimalAttributeMetadata	Range 0.00-1.00; precision 2.
AI Last Run On	aso_ailastrunon	Date and time	DateTimeAttributeMetadata	User-local timestamp.
AI Agent Status	aso_aiaгентstatus	Choice	Global PicklistAttributeMetadata	Uses existing global choice aso_aistatus.
AI SAP Context Summary	aso_aisapcontextsummary	Multiline text	MemoAttributeMetadata	SAP-derived summary after governed integration.
AI SAP Match Flag	aso_aisapmatchflag	Yes/No	BooleanAttributeMetadata	Default No.
AI Correlation ID	aso_aicorrelationid	Text	StringAttributeMetadata	Last orchestration/integration correlation id.
Sales Qualification Agent Status	aso_salesqualificationagentstatus	Choice	Local PicklistAttributeMetadata	Local values: NotStarted, Running, Completed, Failed, NeedsReview.
Sales Qualification Agent Score	aso_salesqualificationagentscore	Whole number	IntegerAttributeMetadata	Range 0-100.
Sales Qualification Agent Rationale	aso_salesqualificationagentrationale	Multiline text	MemoAttributeMetadata	Sales Qualification Agent output.
Sales Qualification Agent Last Run On	aso_salesqualificationagentlastrunon	Date/time	DateTimeAttributeMetadata	User-local timestamp.
Foundry Final Qualification Decision	aso_foundryfinalqualificationdecision	Text	StringAttributeMetadata	MVP text field because taxonomy may evolve.
Foundry Review Required	aso_foundryreviewrequired	Yes/No	BooleanAttributeMetadata	Human review gate; default No.
Communication State	aso_communicationstate	Choice	Global PicklistAttributeMetadata	Uses existing global choice aso_communicationstate.
Lifecycle Communication Stage	aso_lifecyclecommunicationstage	Choice	Global PicklistAttributeMetadata	Uses existing global choice aso_lifecyclecommunicationstage.
Journey Participation Status	aso_journeyparticipationstatus	Choice	Global PicklistAttributeMetadata	Uses existing global choice aso_journeyparticipationstatus.
Customer Insights Journey ID	aso_customerinsightsjourneyid	Text	StringAttributeMetadata	Latest journey reference.
Customer Insights Journey Name	aso_customerinsightsjourneyname	Text	StringAttributeMetadata	Latest journey name.
Customer Insights Last Entry On	aso_customerinsightslastentryon	Date/time	DateTimeAttributeMetadata	Last journey entry timestamp.
Customer Insights Last Interaction On	aso_customerinsightslastinteractionon	Date/time	DateTimeAttributeMetadata	Last Customer Insights interaction timestamp.
Customer Insights Last Interaction Type	aso_customerinsightslastinteractiontype	Choice	Local PicklistAttributeMetadata	Local values: EmailSent, Open, Click, FormSubmit, Reply, Unsubscribe, Bounce, CustomAction.
Email Consent Status	aso_emailconsentstatus	Choice	Global PicklistAttributeMetadata	Uses existing global choice aso_emailconsentstatus.
Compliance Profile Name	aso_complianceprofilename	Text	StringAttributeMetadata	Customer Insights compliance profile name.
Communication Hold Reason	aso_communicationholdreason	Multiline text	MemoAttributeMetadata	Suppression / hold reason.

8. What the script does step by step

1. Defines the target Dataverse environment URL.

2. Defines the target unmanaged solution unique name: ASOCore.
3. Defines the target table logical name: lead.
4. Creates an interactive OAuth connection using ServiceClient.
5. Retrieves existing Lead attributes with RetrieveEntityRequest.
6. Builds a list of desired Lead fields as metadata objects.
7. Loops over the list.
8. Skips a field if its logical name already exists.
9. Creates missing fields with CreateAttributeRequest.
10. Passes SolutionUniqueName = ASOCore so created fields are associated with ASO.Core.
11. Logs each result to the terminal.
12. Publishes Lead table customizations with PublishXmlRequest.

9. Local choices vs global choices

Choice type	Meaning	Example in this run	Why it matters
Global choice	A reusable option set created as its own solution component. Many columns can reference it.	aso_communicationstate	Keeps common states consistent across Lead, Opportunity, Account, and Contact.
Local choice	An option set stored directly on one column.	aso_airoutingdecision	Useful when the values are specific to one field and not yet worth standardizing globally.

10. Terminal commands used

```
# 1. Create and enter the local project folder
mkdir -p ~/aso-dataverse-tools/lead-extension
cd ~/aso-dataverse-tools/lead-extension

# 2. Create a .NET 8 console project
dotnet new console --framework net8.0

# 3. Add the Dataverse client SDK package
dotnet add package Microsoft.PowerPlatform.Dataverse.Client

# 4. Open/edit Program.cs
nano Program.cs

# 5. Run the metadata creation script
dotnet run
```

11. Validation checklist

Check	How to validate	Expected result
Script finished	Terminal output	Done. Validate in ASO.Core -> Tables -> Lead -> Columns.
Fields created	Power Apps -> Phoenicarix-CI -> Solutions -> ASO.Core -> Tables -> Lead -> Columns	New aso_ fields are visible.
No default solution pollution	Open ASO.Core and confirm components are inside the solution	Fields are associated with ASO.Core.
Naming consistency	Search for aso_ in Lead columns	Logical names use aso_ prefix.
Choice reuse	Open AI Agent Status, Communication State, Lifecycle Stage, Journey Participation Status, Email Consent Status	They reference existing global choices.
Local choice values	Open AI Fit Level, AI Routing Decision, Sales Qualification Agent Status, Customer Insights Last Interaction Type	Values match the agreed Phase 2 model.
Publishing	Refresh maker portal after script	Columns are visible and usable.

12. Backup naming after successful Lead extension

Use the agreed backup naming standard: solution, phase, environment, scope, managed/unmanaged marker, version, and date.

ASO.Core_Phase2_CI_LeadExtension_unmanaged_v1_0_20260523.zip
 ASO.Core_Phase2_CI_LeadExtension_managed_v1_0_20260523.zip

13. Troubleshooting

Symptom	Likely cause	Action
ASO.Core not found	Connected to wrong environment or wrong solution unique name	Run pac auth list, select Phoenicarix-CI, run pac solution list grep -i aso.
Global choice error	Choice logical name differs from script	Check ASO.Core -> Choices and adjust GlobalChoice(..., globalChoiceName, ...).
Some fields already exist	Script was rerun or partially completed earlier	Expected; the script skips existing fields.
Authentication window appears	Interactive OAuth login is required	Log in with the same account used in Power Apps.
Field appears in environment but not solution	Solution association failed or wrong solution name	Add existing column to ASO.Core manually, then correct SolutionUniqueName.
Display labels are English in German UI	Script uses language code 1033	Add localized labels later if required.

14. Governance and architecture guardrails

- Dataverse remains the operational sales truth.
- Customer Insights - Journeys remains the only lifecycle communication execution layer.
- The Lead extension prepares fields only; it does not start customer communication.
- SAP fields are references and context fields only; they are not an SAP integration.
- HubSpot fields are ingress references only; HubSpot is not configured in this step.

- AI draft fields are draft-only and must not be sent directly.
- High-impact commercial automation remains out of scope until later phases and approvals exist.

15. Handover notes for customer IT

13. Keep the script in source control, not only on one laptop.
14. Record the run date, environment URL, solution unique name, and script version in the implementation workbook.
15. Export ASO.Core after validation.
16. Do not edit the generated schema names manually after creation.
17. Add the fields to forms only after the schema is validated.
18. Repeat the same pattern for Opportunity, Account, and Contact after Lead is confirmed.

Appendix A - Full script snapshot

This appendix documents the exact script structure used for the Lead extension. In a customer project, store this script in source control and review changes through normal pull-request/code-review workflow.

```
using Microsoft.Crm.Sdk.Messages;
using Microsoft.PowerPlatform.Dataaverse.Client;
using Microsoft.Xrm.Sdk.Messages;
using Microsoft.Xrm.Sdk.Metadata;

const string DataaverseUrl = "https://phoenicarix-ci.crm4.dynamics.com";
const string SolutionUniqueName = "ASOCore";
const string EntityLogicalName = "lead";
const int LanguageCode = 1033;

var connectionString =
    $"AuthType=OAuth;
    Url={DataaverseUrl};
    LoginPrompt=Auto;
    ClientId=51f81489-12ee-4a9e-aaae-a2591f45987d;
    RedirectUri=http://localhost";

using var service = new ServiceClient(connectionString);

if (!service.IsReady)
{
    Console.ForegroundColor = ConsoleColor.Red;
    Console.WriteLine("Connection failed:");
    Console.WriteLine(service.LastError);
    Console.ResetColor();
    return;
}
```

```

Console.WriteLine($"Connected to: {DataverseUrl}");
Console.WriteLine($"Target solution: {SolutionUniqueName}");
Console.WriteLine($"Target table: {EntityLogicalName}");
Console.WriteLine();

var existingAttributes = GetExistingAttributeLogicalNames(service, EntityLogicalName);

var fields = new List<AttributeMetadata>
{
    Text("aso_hubspotcontactid", "HubSpot Contact ID", 100, "External ingress key; do not configure HubSpot yet."),
    Text("aso_hubspotsource", "HubSpot Source", 100, "Ingress source reference; keep governed."),
    Text("aso_sapbusinesspartnerid", "SAP Business Partner ID", 100, "SAP reference only."),
    Text("aso_sapcustomerid", "SAP Customer ID", 100, "SAP reference only."),
    LocalChoice("aso_aifitlevel", "AI Fit Level", new[] { "Strong", "Moderate", "Weak" }, "AI fit level."),
    WholeNumber("aso_aileadscore", "AI Lead Score", 0, 100, "AI lead score 0-100."),
    Memo("aso_aiquaificationrationale", "AI Qualification Rationale", "Seller-facing rationale."),
    Memo("aso_aioutreachdraft", "AI Outreach Draft", "Draft only; not direct-send."),
    LocalChoice("aso_airoutingdecision", "AI Routing Decision", new[] { "Nurture", "SDR", "AE", "Reject", "NeedsReview", "ExistingAccountReview" }, "AI routing decision."),
    DecimalNumber("aso_aiconfidence", "AI Confidence", 0m, 1m, 2, "AI confidence 0.00-1.00."),
    DateTimeField("aso_ailastrunon", "AI Last Run On", "Last AI run timestamp."),
    GlobalChoice("aso_aiaagentstatus", "AI Agent Status", "aso_aistatus", "Uses global AI status choice."),
    Memo("aso_aisapcontextsummary", "AI SAP Context Summary", "SAP-derived summary after governed integration."),
    YesNo("aso_aisapmatchflag", "AI SAP Match Flag", "Likely SAP match."),
    Text("aso_aicorrelationid", "AI Correlation ID", 100, "Last run correlation."),
    LocalChoice("aso_salesqualificationagentstatus", "Sales Qualification Agent Status", new[] { "NotStarted", "Running", "Completed", "Failed", "NeedsReview" }, "Sales qualification agent status."),
    WholeNumber("aso_salesqualificationagentscore", "Sales Qualification Agent Score", 0, 100, "Sales agent score if available."),
    Memo("aso_salesqualificationagentrationale", "Sales Qualification Agent Rationale", "Output from Sales Qualification Agent."),
    DateTimeField("aso_salesqualificationagentlastrunon", "Sales Qualification Agent Last Run On", "Timestamp."),
    Text("aso_foundryfinalqualificationdecision", "Foundry Final Qualification Decision", 100, "Post-validation decision."),
    YesNo("aso_foundryreviewrequired", "Foundry Review Required", "Human review gate."),
    GlobalChoice("aso_communicationstate", "Communication State", "aso_communicationstate", "Uses global Communication State choice."),
    GlobalChoice("aso_lifecyclecommunicationstage", "Lifecycle Communication Stage", "aso_lifecyclecommunicationstage", "Uses global Lifecycle Communication Stage choice."),
    GlobalChoice("aso_journeyparticipationstatus", "Journey Participation Status", "aso_journeyparticipationstatus", "Uses global Journey Participation Status choice."),
    Text("aso_customerinsightsjourneyid", "Customer Insights Journey ID", 100, "Latest journey reference."),
    Text("aso_customerinsightsjourneyname", "Customer Insights Journey Name", 200, "Latest journey name."),
    DateTimeField("aso_customerinsightslastentryon", "Customer Insights Last Entry On", "Last entry."),
    DateTimeField("aso_customerinsightslastinteractionon", "Customer Insights Last Interaction On", "Last interaction."),
    LocalChoice("aso_customerinsightslastinteractiontype", "Customer Insights Last Interaction Type", new[] { "EmailSent", "Open", "Click", "FormSubmit", "Reply", "Unsubscribe", "Bounce", "CustomAction" }, "Customer Insights last interaction type."),
    GlobalChoice("aso_emailconsentstatus", "Email Consent Status", "aso_emailconsentstatus", "Uses global Email Consent Status choice."),
    Text("aso_complianceprofilename", "Compliance Profile Name", 200, "Profile used."),
    Memo("aso_communicationholdreason", "Communication Hold Reason", "Suppression / hold reason.")
};

foreach (var field in fields)
{
    var logicalName = field.SchemaName!.ToLowerInvariant();

    if (existingAttributes.Contains(logicalName))
    {
        Console.ForegroundColor = ConsoleColor.Yellow;
    }
}

```



```

        Console.WriteLine($"SKIP existing: {logicalName}");
        Console.ResetColor();
        continue;
    }

    try
    {
        Console.WriteLine($"Creating: {logicalName} ...");
        service.Execute(new CreateAttributeRequest
        {
            EntityName = EntityLogicalName,
            Attribute = field,
            SolutionUniqueName = SolutionUniqueName
        });
        Console.ForegroundColor = ConsoleColor.Green;
        Console.WriteLine($"CREATED: {logicalName}");
        Console.ResetColor();
    }
    catch (Exception ex)
    {
        Console.ForegroundColor = ConsoleColor.Red;
        Console.WriteLine($"FAILED: {logicalName}");
        Console.WriteLine(ex.Message);
        Console.ResetColor();
    }
}

Console.WriteLine();
Console.WriteLine("Publishing Lead table customizations...");
service.Execute(new PublishXmlRequest
{
    ParameterXml = "<importexportxml><entities><entity>lead</entity></entities></importexportxml>"
});
Console.ForegroundColor = ConsoleColor.Green;
Console.WriteLine("Done. Validate in ASO.Core -> Tables -> Lead -> Columns.");
Console.ResetColor();

static HashSet<string> GetExistingAttributeLogicalNames(ServiceClient service, string entityLogicalName)
{
    var response = (RetrieveEntityResponse)service.Execute(new RetrieveEntityRequest
    {
        LogicalName = entityLogicalName,
        EntityFilters = EntityFilters.Attributes,
        RetrieveAsIfPublished = true
    });
    return response.EntityMetadata.Attributes
        .Where(a => !string.IsNullOrEmpty(a.LogicalName))
        .Select(a => a.LogicalName!)
        .ToHashSet(StringComparer.OrdinalIgnoreCase);
}

static Label Label(string value) => new(value, LanguageCode);
static AttributeRequiredLevelManagedProperty Optional() => new(AttributeRequiredLevel.None);

static StringAttributeMetadata Text(string schemaName, string displayName, int maxLength, string description) => new()

```

```

{
    SchemaName = schemaName,
    DisplayName = Label(displayName),
    Description = Label(description),
    RequiredLevel = Optional(),
    MaxLength = maxLength
};

static MemoAttributeMetadata Memo(string schemaName, string displayName, string description) => new()
{
    SchemaName = schemaName,
    DisplayName = Label(displayName),
    Description = Label(description),
    RequiredLevel = Optional(),
    MaxLength = 4000
};

static IntegerAttributeMetadata WholeNumber(string schemaName, string displayName, int min, int max, string description) => new()
{
    SchemaName = schemaName,
    DisplayName = Label(displayName),
    Description = Label(description),
    RequiredLevel = Optional(),
    MinValue = min,
    MaxValue = max,
    Format = IntegerFormat.None
};

static DecimalAttributeMetadata DecimalNumber(string schemaName, string displayName, decimal min, decimal max, int precision, string description)
=> new()
{
    SchemaName = schemaName,
    DisplayName = Label(displayName),
    Description = Label(description),
    RequiredLevel = Optional(),
    MinValue = min,
    MaxValue = max,
    Precision = precision
};

static DateTimeAttributeMetadata DateTimeField(string schemaName, string displayName, string description) => new()
{
    SchemaName = schemaName,
    DisplayName = Label(displayName),
    Description = Label(description),
    RequiredLevel = Optional(),
    Format = DateTimeFormat.DateAndTime,
    DateTimeBehavior = DateTimeBehavior.UserLocal
};

static BooleanAttributeMetadata YesNo(string schemaName, string displayName, string description) => new()
{
    SchemaName = schemaName,
    DisplayName = Label(displayName),
    Description = Label(description),

```

```

        RequiredLevel = Optional(),
        DefaultValue = false,
        OptionSet = new BooleanOptionSetMetadata(
            new OptionMetadata(Label("Yes"), 1),
            new OptionMetadata(Label("No"), 0))
    };

    static PicklistAttributeMetadata LocalChoice(string schemaName, string displayName, string[] values, string description)
    {
        var optionSet = new OptionSetMetadata { IsGlobal = false, OptionSetType = OptionSetType.Picklist };
        foreach (var value in values)
        {
            optionSet.Options.Add(new OptionMetadata(Label(value), null));
        }
        return new PicklistAttributeMetadata
        {
            SchemaName = schemaName,
            DisplayName = Label(displayName),
            Description = Label(description),
            RequiredLevel = Optional(),
            OptionSet = optionSet
        };
    }

    static PicklistAttributeMetadata GlobalChoice(string schemaName, string displayName, string globalChoiceName, string description) => new()
    {
        SchemaName = schemaName,
        DisplayName = Label(displayName),
        Description = Label(description),
        RequiredLevel = Optional(),
        OptionSet = new OptionSetMetadata { IsGlobal = true, Name = globalChoiceName, OptionSetType = OptionSetType.Picklist }
    };

```